DESIGN and SIMULANTION of an AUTONOMOUS RIVER CLEANING ROBOT

Haoxuan Li, Yifei Shen, Yihan Zhou, Jianing Wang, Changbo Zheng Supervisor: Dr Jieming Ma

Abstract

Currently, the amount of river waste in China is increasing, and the surface waste in rivers and tributaries, if not treated, can cause significant damage to the environment. This project developed an intelligent solar multi-rotor surface debris collection system. In this project, we built a 3D model in Rhino and imported it into ROS to simulate the robot's working environment for testing. Through the experiment, we can conclude that the garbage collection system can collect more than 90% of the garbage.

Introduction

Most of the current river cleaning techniques in China are manual. The disadvantages are unsatisfactory cleaning results and high labor costs. The industry has adopted various automated cleaning tools to solve these problems, namely using motorized boats to collect debris from the river. However, current technology has problems, including inefficiency, excessive energy consumption, and impact on the hydrological environment. In addition, conventional gear and equipment use large amounts of energy and, over their lifetime, use large amounts of water [1]. An autonomous photovoltaic water floating collector has been developed to address these issues. The innovation aims to create clean solar energy using vessels designed to remove surface debris from a specific watershed. ROS Gazebo was chosen as our modelling tool because it supports various programming languages, physics engines, and model types [2]. This turntable-based synergistic collection method is an improvement over the single cleaner collection method currently in use. The collection of floating material is substantially improved as it passes through the motor-driven turntable set, which moves in the direction of the waste collection cage as the blades rotate.

Figure 1. River Pollution Problem

Method

1) Modelling

In the research, this 3D model was constructed in Rhino and imported into Robot Operating System (ROS) Gazebo to build static models. An environment is set up to simulate the working conditions of the robot. The gear can rotate on the water surface by editing the code to achieve its function. We can also control the flow of the river in the simulation program.

2) Principle

The photovoltaic solar panels above the turntable and the propeller design of the turntable's main body can fully use solar energy to charge the motor, thus giving power to the turntable group. Even in conditions with low sunlight, the robot can use wind power to rotate. When the floating objects pass the turntable set, driven by the motor, the floating debris move in the direction of the garbage collection cage with the rotation of the blades, thus realizing the garbage collection.

3) Workflow

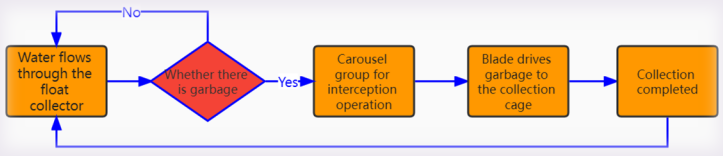

Figure 2. Flowchart of Robot's Working Process

Result

As shown in the Figure 3, a collection of rotors forms a line to pass floating debris in the river into the collection basket. The rotors catch the debris on the water's surface through extended leaf blades. As the rotors spin, debris is passed from one rotor to another and eventually into the collection basket. The blue plate on top of the rotor is a solar cell that provides electricity to power a small motor that turns each rotor. The wind turbines on the body of the rotor can power the rotor using energy from the wind.

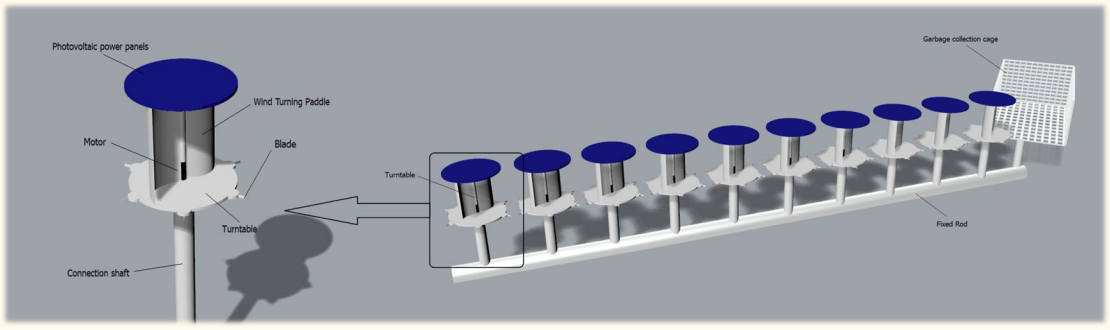

Figure 3. Surface Debris Collection System

Our simulation setup is shown in Fig. 4. Debris of various sizes are set up to flow through our robot. The simulation can control how fast the river flows and how quickly each rotor rotates. The robot's effectiveness is determined by the percentage of debris successfully collected. The results are shown in the data table below.

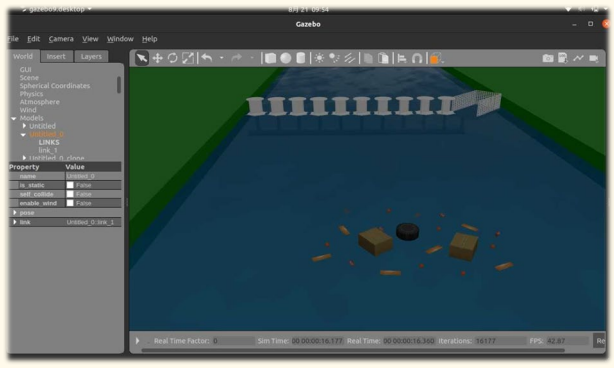

Figure 4. Gazebo Simulation Environment

River Flow Speed	Rotor Turn Speed	Collected	Missed	Efficiency
Fast	Slow	16	4	80%
Fast	Fast	18	2	90%
Slow	Fast	20	0	100%
Slow	Slow	19	1	95%

Figure 5. Simulation Results

Conclusion and Future Development

The result from the Gazebo simulation is promising because our debris collection system effectively collected an average of 91.25% of the river debris. The simulation results show that collecting trash is more challenging if the river flow rate is fast. Also, the rotor rotation speed is proportional to the collection efficiency. At present, the Patent Office is reviewing the invention patent of our design. In the future, we would like to 3D print our designed robot to use in real life.

References:

[1] Q. Yang, T. Huang, F. Chen, J. Uche, Y. Wang, P. Yuan, Y. Zhang, J. Li, 'Water saving potential for large-scale photovoltaic power generation in China: Based on life cycle assessment', Renewable and Sustainable Energy Reviews, vol.167, no.112681, pp.1-15,2022.

[2] A. Farley, J. Wang, J A. Marshall, 'How to pick a mobile robot simulator: A quantitative comparison of CoppeliaSim, Gazebo, MORSE and Webots with a focus on accuracy of motion', Simulation Modelling Practice and Theory, vol.120, no.102629, pp, 1-16,2022